

Lecture 1 — Pricing with Full Information

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1.1 Why are pricing problems difficult?

Two major sources of difficulty will recur throughout the course:

- **Uncertainty about future events:** future demand, arrivals, capacity needs, and market conditions are not known when a decision is made.
- **Private information:** buyers may know their values and preferences, or sellers their costs, while the platform or trading partner does not.

For today, we deliberately remove both difficulties. Suppose all relevant values and costs are known. What is left for a price to do?

Even under full information, prices can divide gains from trade, help allocate scarce supply, coordinate decentralized choices across heterogeneous goods, manage dynamic opportunity costs, and internalize congestion externalities. This leads to a course-level organizing question:

When do simple prices coordinate a market, and when do we need richer rules?

1.2 One buyer and one seller: dividing the gains from trade

There is one buyer, one seller, and one indivisible item. The buyer values the item at v , the seller has cost or opportunity cost c , and the posted price is p . Write

$$x^+ = \max\{x, 0\}.$$

Relative to the outside option of not trading, the two sides' individual participation payoffs are

$$U_B(v, p) = (v - p)^+, \quad U_S(c, p) = (p - c)^+. \quad (1.1)$$

The positive-part notation records voluntary participation: neither side is forced into a trade that gives negative utility. Actual trade requires both sides to accept, so a mutually acceptable price lies in $[c, v]$.

If $p \in [c, v]$, trade occurs and

$$\underbrace{v - c}_{\text{total gains from trade}} = \underbrace{v - p}_{\text{buyer surplus}} + \underbrace{p - c}_{\text{seller surplus}}.$$

This separates two questions:

Efficiency asks whether trade should occur; price determines how the gains from trade are divided.

Efficient trade occurs when $v \geq c$. If $v < c$, the total gains from trade are negative, so efficient trade does not occur.

Now specialize to seller revenue with zero cost. For a nonnegative posted price and the convention that an indifferent buyer accepts,

$$R(p) = p \mathbf{1}\{p \leq v\}. \quad (1.2)$$

Under full information,

$$p^* = v.$$

The seller extracts all buyer surplus only because the value v is known. Once values become private information, charging each buyer exactly their value is no longer an available rule; that difficulty motivates much of the later course.

1.3 Many buyers and identical items

Suppose there are n unit-demand buyers with known values

$$v_{(1)} \geq v_{(2)} \geq \cdots \geq v_{(n)},$$

and one seller with k identical items and zero cost. Although the values are known, the seller is restricted to a *single anonymous price*: every buyer faces the same price, so the seller cannot price discriminate.

First let $k = 1$. The revenue-maximizing price is

$$p^* = v_{(1)}.$$

The highest-valued buyer receives the item. Revenue maximization and welfare maximization happen to align in this simplest case.

What changes when $k > 1$? Consider the classroom example

$$k = 2, \quad \text{values } 10, 4, 3, 2.$$

At price $p = 4$, two units sell. The market clears, welfare is

$$10 + 4 = 14,$$

and revenue is $2 \cdot 4 = 8$. At price $p = 10$, only one unit sells and revenue is 10; the second unit remains unused. Therefore

revenue maximization need not clear the market.

In general, if price $v_{(j)}$ sells to the top j buyers, the best anonymous-price revenue is

$$R^* = \max_{1 \leq j \leq k} j v_{(j)}.$$

Ignoring ties, a price that induces exactly k buyers to demand the good satisfies

$$v_{(k+1)} < p \leq v_{(k)}.$$

Such a price clears the market. The phrase “optimal price” is incomplete until we specify the objective: welfare, market clearing, or revenue.

1.4 Heterogeneous goods: relative prices coordinate choices

With heterogeneous items, a buyer’s demand for one item depends on the prices of all the others. Consider two unit-demand buyers and two items A, B , with valuation matrix

$$V = \begin{pmatrix} 10 & 6 \\ 8 & 7 \end{pmatrix}.$$

Rows are buyers and columns are items. The efficient assignment is

$$1 \rightarrow A, \quad 2 \rightarrow B,$$

with welfare $10 + 7 = 17$.

Start from zero prices, $p_A = p_B = 0$. Both buyers prefer item A , so their decentralized choices do not coordinate the efficient allocation. How much more expensive should A be than B so that buyer 2 switches to B , while buyer 1 still prefers A ?

Buyer 1 prefers A when

$$10 - p_A \geq 6 - p_B \quad \Longleftrightarrow \quad p_A - p_B \leq 4.$$

Buyer 2 prefers B when

$$7 - p_B \geq 8 - p_A \quad \Longleftrightarrow \quad p_A - p_B \geq 1.$$

Hence any relative price difference satisfying

$$\boxed{1 \leq p_A - p_B \leq 4} \quad (1.3)$$

coordinates the efficient assignment.

For example, take $p_A = 3$ and $p_B = 1$. The buyers' net utilities are

	A	B
1	7	5
2	5	6

Buyer 1 chooses A and buyer 2 chooses B . The economic point comes before the optimization machinery:

Prices coordinate decentralized choices by making the more contested item relatively more expensive.

1.5 The assignment LP and dual prices

The centralized welfare-maximizing assignment for this 2×2 example is

$$\begin{aligned} \max_{x \geq 0} \quad & \sum_{i \in \{1,2\}} \sum_{j \in \{A,B\}} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j \in \{A,B\}} x_{ij} \leq 1 & i = 1, 2, \\ & \sum_{i \in \{1,2\}} x_{ij} \leq 1 & j = A, B. \end{aligned}$$

Here x_{ij} indicates whether buyer i receives item j . The assignment polytope is integral, so solving the LP relaxation produces an actual assignment.

Associate buyer variables $u_i \geq 0$ with the first constraints and item variables $p_j \geq 0$ with the second. The dual is

$$\begin{aligned} \min_{u, p \geq 0} \quad & \sum_i u_i + \sum_j p_j \\ \text{s.t.} \quad & u_i + p_j \geq v_{ij} & \forall i, j. \end{aligned}$$

Interpreting p_j as the price of item j and u_i as buyer i 's utility, dual feasibility says

$$u_i + p_j \geq v_{ij} \iff u_i \geq v_{ij} - p_j. \quad (1.4)$$

Thus u_i is at least the net utility buyer i could obtain from every item.

The coordinating prices $p_A = 3, p_B = 1$ give utilities $u_1 = 7, u_2 = 6$. They are dual feasible, and the dual objective is

$$7 + 6 + 3 + 1 = 17,$$

equal to the primal welfare. Hence both solutions are optimal.

In general, complementary slackness says that on an allocated edge,

$$u_i = v_{ij} - p_j,$$

while dual feasibility ensures that no other item offers buyer i greater net utility. Therefore optimal item prices can support the efficient assignment through decentralized choice:

LP dual variables can become market prices.

Remark 1.1 (Generalization and Assignment 1). *For m buyers and n items, the same primal has one constraint per buyer and item, and the same dual has inequalities $u_i + p_j \geq v_{ij}$ for every pair. Assignment 1 develops the full primal–dual and complementary-slackness argument and applies it to the existing 3×3 valuation matrix. Supporting prices need not be unique, and prices chosen to support efficiency are not automatically revenue-maximizing prices.*

1.6 Queue Lab and a Naor-style queue

We now go back to a single homogeneous service. Surprisingly, pricing can again become subtle even though every customer's value is known and identical.

The classical Naor model studies strategic entry into a queue. Here we use two deliberately simple discrete-time variants. A deterministic calibration first shows why extracting as much as possible from the *current* customer can be a poor long-run policy. We then turn to stochastic arrivals to isolate the congestion externality.

Remark 1.2 (Companion activity: Queue Lab). *The companion Queue Lab game separates two objectives that should not be conflated.*

- (1) *In Calibration Day, price the first few scheduled robots manually and then program a workload-dependent controller. The score is operator credits per cycle.*
- (2) *In Quiet Shift and Rush Hour, replay a fixed dispatch under different controllers. The score is city surplus per request.*
- (3) *Keep both operator credits and city surplus visible. A controller that performs well for one objective need not perform well for the other.*

- (4) The optional Mixed Fleet mode varies values, waiting costs, and repair requirements; it asks whether workload alone remains a sufficient state for pricing.

The calculations below explain the deterministic calibration and the two homogeneous stochastic shifts.

1.6.1 Calibration Day: current extraction versus long-run revenue

In Calibration Day, one robot arrives every two cycles. Each repair requires $s = 3$ cycles, every robot has value $V = 5$, and waiting costs $c = 1$ per cycle. If W_n is the workload seen by the n th scheduled robot and J_n indicates whether it docks, then

$$W_{n+1} = [W_n + 3J_n - 2]^+.$$

A workload-dependent access price $\tau(w)$ gives the arriving robot utility

$$U(w) = 5 - \tau(w) - w.$$

Suppose the operator behaves myopically and extracts all of the current robot's residual willingness to pay whenever possible:

$$\tau_{\text{myopic}}(w) = [5 - w]^+.$$

Starting from $W_1 = 0$, the first six robots dock at workloads $0, 1, \dots, 5$ and pay $5, 4, \dots, 0$. The workload then cycles through $4, 5, 6$: two robots dock, one reroutes, and the operator collects only one credit every six cycles.

By contrast, the fixed price $\tau = 4$ produces the workload cycle $0, 1, 2$: two robots dock at price 4, then one reroutes while the bay drains. It collects eight credits every six cycles. Thus

$$\underbrace{\text{extract the most from the next customer}}_{\text{myopic rule}} \neq \underbrace{\text{maximize revenue over time}}_{\text{dynamic objective}}.$$

The point is not that a fixed price is always optimal. It is that admitting work changes the state faced by future customers, so a pricing policy must account for an opportunity cost that a one-customer calculation misses. This is an early preview of dynamic pricing and revenue management.

1.6.2 The stochastic workload model

Time is slotted, $t = 0, 1, 2, \dots$. There is one server. Let

$$W_t \in \mathbb{Z}_+$$

be the unfinished workload (in service slots) immediately before the arrival decision in period t .

In every period a potential customer arrives independently with probability λ . Every admitted customer requires exactly

$$s = 4$$

service slots. The value of service is

$$V = 10,$$

and waiting costs one unit per slot. If the platform charges an admission toll $\tau(w)$, a customer arriving to workload $W_t = w$ gets utility

$$U(w) = 10 - \tau(w) - w.$$

Hence a self-interested customer joins exactly when

$$w \leq 10 - \tau(w).$$

If $J_t \in \{0, 1\}$ indicates that an arrival is admitted, workload evolves as

$$W_{t+1} = [W_t + sJ_t - 1]^+.$$

The toll therefore acts as an admission-control instrument. A fixed toll τ creates the workload threshold $w \leq 10 - \tau$; a programmed controller $\tau(w)$ permits richer state-dependent admission rules.

The game reports normalized versions of two different accounting measures. Operator credits are the toll payments,

$$\sum_t \tau(W_t) J_t.$$

City surplus counts the value of each admitted repair minus that robot's waiting cost,

$$\sum_t (10 - W_t) J_t.$$

The toll is a transfer from the robot to the operator, so it is not subtracted from city surplus. More importantly, neither expression by itself displays the future delay that today's admission creates; that effect appears through later workloads.

1.6.3 The load parameter

First imagine that all potential arrivals are admitted. When the server is busy,

$$\mathbb{E}[W_{t+1} - W_t] \approx s\lambda - 1.$$

This identifies the familiar load parameter

$$\rho = s\lambda.$$

The sign of $\rho - 1$ already tells us something important.

1.6.4 Example 1: low congestion

Take

$$\lambda = 0.05, \quad s = 4,$$

so

$$\rho = s\lambda = 0.2 \ll 1.$$

When the server is busy, expected workload drift under unrestricted admission is

$$0.2 - 1 = -0.8.$$

Thus workload tends to drain rapidly.

We can also make a simple local externality calculation. Suppose the system is initially empty and a customer enters. A later arrival r slots into this customer's service is delayed by roughly $(s - r)^+$ slots solely because of this customer. Ignoring downstream interactions, the expected direct waiting cost imposed on arrivals during these four service slots is

$$\lambda \sum_{r=1}^{s-1} (s - r) = 0.05(3 + 2 + 1) = 0.3.$$

Relative to the service value $V = 10$, the direct congestion externality is small in this low-load example.

1.6.5 Example 2: congestion

Now take

$$\lambda = 0.4, \quad s = 4,$$

so

$$\rho = s\lambda = 1.6 > 1.$$

If everybody were admitted, then while the system is busy,

$$\mathbb{E}[W_{t+1} - W_t] \approx 1.6 - 1 = 0.6 > 0,$$

so workload would grow rather than drain. Some form of admission control is essential for a stable, usable system.

The same local externality calculation gives

$$\lambda \sum_{r=1}^{s-1} (s - r) = 0.4(3 + 2 + 1) = 2.4,$$

even before accounting for downstream congestion effects. Yet an entering customer considers only their own waiting cost w , not the additional delay imposed on other customers.

At zero toll, the private entry rule is

$$w \leq 10.$$

A positive fixed toll shifts this threshold downward. For example, $\tau = 4$ produces the private rule

$$w \leq 6.$$

We are *not* claiming that 6 is the socially optimal threshold in this example; the point is that any desired threshold $h \leq 10$ can be decentralized by the toll

$$\tau = 10 - h.$$

Thus, if a planner accounts for congestion and chooses a social admission threshold h^* below the laissez-faire threshold, a price can implement it.

Remark 1.3 (The Naor lesson). *The customer internalizes their own waiting cost but not the waiting cost their entry creates for others. Thus private entry can exceed socially desirable entry. Here price is not merely dividing surplus or rationing a fixed stock of goods: it is an instrument for internalizing a congestion externality.*

The two parts of Queue Lab therefore expose two distinct reasons to condition current admission on future consequences. Under a revenue objective, accepting a low-paying robot may crowd out more valuable future access. Under a welfare objective, an entering robot may impose delay on others. In both cases the state variable W_t compresses the relevant history, but the correct controller depends on the objective.

1.7 What did “price” mean in these examples?

The examples can be summarized as follows.

Setting	Role of price
One buyer, one item	Divide gains from trade; under full information, extract buyer surplus.
Identical goods	Allocate scarce supply; market clearing and revenue maximization need not coincide.
Heterogeneous goods	Coordinate substitution and decentralized choices; LP dual variables act as item prices.
Congested service	Manage dynamic opportunity cost under a revenue objective; internalize a delay externality under a welfare objective.

The overarching lesson is that there is no unique notion of the “right” price without specifying both the objective and the institutional environment.

1.8 Looking ahead

A recurring question: what changes with scale?

Large systems can be harder because their state and action spaces defeat exact optimization. Yet scale can also create useful regularity: aggregate quantities stabilize, competition and matching opportunities thicken, and individual participants may become less pivotal. The queue's load parameter $\rho = s\lambda$ is our first glimpse of this compression—many arrival and service events summarized by one first-order quantity. We will keep asking which benefit or cost of scale is operating.

Everything in this lecture assumed that values were known. In most real markets they are not. The next step is therefore to replace the known numbers v_i by a model of heterogeneous values,

$$V \sim F,$$

and understand how a distribution of values generates a demand curve

$$D(p) = N \Pr(V \geq p) = N(1 - F(p)).$$

This leads to the next question: *where does demand come from, and what can we infer about it from data?*

Recommended reading

Core / closest to this lecture.

Vohra and Krishnamurthi, *Principles of Pricing*, Chapter 1 and Sections 2.1–2.2. Read for the one-buyer surplus calculation, outside-option reasoning, and the basic full-information pricing setup.

Assignment companion.

Assignment 1 is the place for the full general assignment-LP, duality, and complementary-slackness derivation, including the 3×3 numerical instance introduced in class. It develops in detail why dual variables can support market-clearing prices.

Forward-looking alternative.

Talluri and van Ryzin, *The Theory and Practice of Revenue Management*, Sections 2.1.1–2.1.2, introduces capacity controls, displacement cost, and the opportunity-cost viewpoint that becomes central in Lectures 5–6.

Optional / historical.

Naor, “The Regulation of Queue Size by Levying Tolls” (1969), studies the classical queue-entry model and welfare tolls. Queue Lab is a pedagogical variant; the full analysis is not expected.